MODERN PHYSICS

Mock Test 3

Time: 50 minutes Maximum marks: 32

Instructions. This paper has 32 multiple-choice questions. Each question has exactly one correct option and carries 1 mark. Mark your answers on the OMR sheet.

Data. $h = 6.63 \times 10^{-34}\,\mathrm{J\,s}$, $\hbar = 1.055 \times 10^{-34}\,\mathrm{J\,s}$, $c = 3 \times 10^{8}\,\mathrm{m/s}$, $hc = 1240\,\mathrm{eV\,nm}$, $1\,\mathrm{eV} = 1.6 \times 10^{-19}\,\mathrm{J}$, $e = 1.6 \times 10^{-19}\,\mathrm{C}$, $m_e = 9.11 \times 10^{-31}\,\mathrm{kg}$, $R = 1.097 \times 10^{7}\,\mathrm{m^{-1}}$, $a_0 = 0.53\,\text{Å}$, Bohr magneton $\mu_B = \dfrac{e\hbar}{2m_e} = 9.27 \times 10^{-24}\,\mathrm{J/T}$.

Q1. In vacuum the speed of electromagnetic waves is $c = 1/\sqrt{\mu_0 \varepsilon_0}$. In a hypothetical universe in which ε_0 is four times larger and μ_0 is unchanged, the speed of light would be

(A) $c/4$ (B) $c/2$ (C) $2c$ (D) $4c$

Q2. Every object above absolute zero emits electromagnetic radiation. The reason is that

(A) its vibrating atoms and electrons are accelerating charges, and accelerating charges radiate

(B) photons are stored inside the object and slowly leak out

(C) only the free electrons in metals are able to radiate

(D) radiation is emitted only when the object is hot enough to glow visibly

Q3. As the temperature of a hot object is raised, its colour changes from red towards bluish-white. In the observed spectrum, raising the temperature

(A) moves the peak to lower frequency and lowers the intensity

(B) keeps the peak fixed and raises the intensity equally at all frequencies

(C) moves the peak to lower frequency and raises the intensity

(D) moves the peak to higher frequency and raises the intensity

Q4. Light of photon energy 3.1 eV and power 1.0 mW falls on a photocathode whose work function is below 3.1 eV. Exactly 1.0% of the incident photons eject an electron, and all emitted electrons are collected. The saturation photocurrent is about

(A) 0.32 μA (B) 32 μA (C) 3.2 μA (D) 320 μA

Q5. Electrons are sent one at a time towards a double slit, and a fringe pattern builds up on the detector. At the position of a dark fringe, the detection probability is zero with both slits open. If one slit is now closed, the probability of detecting an electron at that position

(A) remains zero, since fewer electrons pass through the slits

(B) becomes nonzero, because the interference that cancelled the amplitude there is no longer present

(C) becomes zero at every position on the screen

(D) is unchanged, because each electron passes through only one slit

Q6. In an X-ray tube operated at a fixed voltage, the tube current is doubled. Which of the following describes the change in the X-ray spectrum?

(A) The intensity doubles and the cutoff wavelength $\lambda_{\min}$ halves.

(B) The intensity doubles and the cutoff wavelength $\lambda_{\min}$ is unchanged.

(C) The intensity is unchanged and the cutoff wavelength $\lambda_{\min}$ halves.

(D) The intensity and the cutoff wavelength $\lambda_{\min}$ are both unchanged.

Q7. Electrons accelerated through 54 V have a de Broglie wavelength of about 1.67 Å. The accelerating voltage needed to give a wavelength of 0.83 Å is

(A) 27 V (B) 108 V (C) 216 V (D) 432 V

Q8. A proton and an alpha particle ($m_\alpha = 4m_p$) have the same kinetic energy. The ratio λ_p/λ_α of their de Broglie wavelengths is

(A) 1/4 (B) 1/2 (C) 1 (D) 2

Q9. The uncertainty principle $\Delta x\,\Delta p \geq \hbar/2$ arises because

(A) real measuring instruments are imperfect, and better instruments could in principle remove the limit

(B) every measurement of position unavoidably disturbs the particle through clumsy apparatus

(C) a particle is described by a wave, so it cannot have both a sharply defined position and a sharply defined momentum

(D) electrons move too fast for their positions to be photographed

Q10. The equation $i\hbar\,\dfrac{\partial \psi}{\partial t} = \hat{H}\psi$ is used to

(A) find how ψ changes with time once the initial state $\psi(\vec{r},0)$ is specified

(B) find the exact trajectory $\vec{r}(t)$ of the electron

(C) normalize the wavefunction

(D) find the allowed energies only, with no information about time dependence

Q11. A free electron ($V = 0$) is in the state $\psi(x) = \sin(kx)$ with $k = 1.0 \times 10^{10}$ m^{-1}. Using $\hat{H} = -\dfrac{\hbar^2}{2m}\dfrac{d^2}{dx^2}$, the energy eigenvalue is about

(A) 0.38 eV (B) 3.8 eV (C) 7.6 eV (D) 150 eV

Q12. A hydrogen atom is placed in a magnetic field of 1.0 T. The energy separation between adjacent Zeeman sublevels of a given l is about

(A) 5.8×10^{-5} eV (B) 1.2×10^{-4} eV (C) 5.8×10^{-3} eV (D) 9.3×10^{-24} eV

Q13. The orbital magnetic moment of an electron is $\vec{\mu} = -\dfrac{e}{2m_e}\vec{L}$. For an electron with $l = 1$, the magnitude of $\vec{\mu}$ is

(A) $\mu_B/2$ (B) μ_B (C) $2\mu_B$ (D) $\sqrt{2}\,\mu_B$

Q14. Using the selection rules $\Delta l = \pm 1$ and $\Delta m_l = 0, \pm 1$, which of the following hydrogen transitions is allowed?

(A) $2s \to 1s$ (B) $3p \to 2p$ (C) $4f \to 3s$ (D) $3d \to 2p$

Q15. For a hydrogen-like ion, $E_n = -13.6\,Z^2/n^2$ eV. In a many-electron atom the other electrons screen the nuclear charge, and the levels are approximated by the same formula with

(A) Z replaced by an effective charge $Z_e < Z$; an $L \to K$ photon is an X-ray only for atoms with large enough Z

(B) Z replaced by an effective charge $Z_e > Z$; an $L \to K$ photon is an X-ray only for atoms with large enough Z

(C) Z replaced by an effective charge $Z_e < Z$; an $L \to K$ photon is an X-ray for every atom, including hydrogen

(D) Z unchanged, because inner electrons do not interact with each other

Q16. In Thomson's plum-pudding model the positive charge is spread through the whole atom and the light electrons are embedded in it. Alpha particles passing through a thin gold foil would then be expected to be

(A) scattered backward through large angles

(B) deflected only through small angles

(C) absorbed by the foil

(D) scattered preferentially through 90°

Q17. Classical electromagnetism predicts that an electron orbiting a nucleus radiates at its orbital frequency and loses energy. The predicted behaviour of the atom is

(A) a continuous spectrum, with the orbit shrinking until the electron falls into the nucleus

(B) a discrete set of lines at the orbital frequency, with a stable atom

(C) no radiation at all, with a stable atom

(D) discrete lines at the observed Balmer wavelengths

Q18. The longest wavelength in the Lyman series of hydrogen ($n \to 1$) is about

(A) 91.2 nm (B) 97.3 nm (C) 121.5 nm (D) 656 nm

Q19. The shortest wavelength in the Brackett series of hydrogen ($n \to 4$) is about

(A) 820 nm (B) 1459 nm (C) 2280 nm (D) 4051 nm

Q20. In the Bohr model the circumference of an allowed orbit holds a whole number of de Broglie wavelengths. The de Broglie wavelength of the electron in the $n = 2$ orbit of hydrogen is about

(A) 3.3 Å (B) 2.1 Å (C) 13.3 Å (D) 6.7 Å

Q21. Which of the following is a limitation of Bohr's model of hydrogen?

(A) It gives the wrong energy levels for hydrogen.

(B) It assigns the ground state an angular momentum of $\hbar$, whereas the true ground state ($1s$) has $l = 0$.

(C) It predicts that the energy levels are continuous.

(D) It cannot account for the existence of the Balmer series.

Q22. In the Bohr model, the kinetic energy of the electron in the $n = 2$ level of hydrogen is

(A) 3.4 eV (B) 6.8 eV (C) 1.7 eV (D) 13.6 eV

Q23. Which of the following functions can **not** be an acceptable wavefunction of a particle on the whole x-axis?

(A) e^{-x^2} (B) $x\,e^{-x^2}$ (C) e^{x^2} (D) $\dfrac{1}{1+x^2}$

Q24. A particle is in the ground state of an infinite well of width L, with $\psi = \sqrt{2/L}\,\sin(\pi x/L)$. The probability of finding it in the central half of the well, $L/4 < x < 3L/4$, is about

(A) 0.50 (B) 0.64 (C) 0.71 (D) 0.82

Q25. The number of nodes (points where $\psi = 0$, excluding $x \to \pm\infty$) of the harmonic-oscillator eigenfunction with $n = 3$ is

(A) 1 (B) 2 (C) 3 (D) 4

Q26. The ground-state energy of an electron in an infinite well of width 0.50 nm is

(A) 0.38 eV (B) 0.75 eV (C) 1.5 eV (D) 6.0 eV

Q27. For a particle in a finite square well, the penetration depth into the classically forbidden region is

(A) the same for all energy states

(B) larger for higher n, since $U - E$ becomes smaller

(C) smaller for higher n, since the particle is more strongly confined

(D) zero for the ground state

Q28. When the Schrödinger equation for hydrogen is separated as $\psi = R(r)\Theta(\theta)\Phi(\phi)$, the potential energy $U(r) = -e^2/4\pi\varepsilon_0 r$ appears in

(A) the azimuthal equation for $\Phi(\phi)$ only

(B) the polar equation for $\Theta(\theta)$ only

(C) all three equations

(D) the radial equation for $R(r)$ only

Q29. The radial equation of hydrogen contains the centrifugal term $\dfrac{l(l+1)\hbar^2}{2m_e r^2}$. At the same r, the ratio of this term for $l = 2$ to its value for $l = 1$ is

(A) 2 (B) 3 (C) 4 (D) 6

Q30. Which of the following hydrogen state labels is **not** allowed?

(A) $2d$ (B) $3d$ (C) $4f$ (D) $1s$

Q31. The number of maxima in the radial probability distribution $P(r)$ of the hydrogen $3s$ state is

(A) 1 (B) 2 (C) 3 (D) 4

Q32. An electron in hydrogen is described by a superposition of two stationary states of energies $E_m > E_n$. Its expectation value $\langle x \rangle$ oscillates in time, and the atom radiates at frequency

(A) $(E_m + E_n)/h$ (B) E_m/h (C) $(E_m - E_n)/2h$ (D) $(E_m - E_n)/h$

End of question paper

Answer Key and Solutions

1	2	3	4	5	6	7	8
B	A	D	C	B	B	C	D

9	10	11	12	13	14	15	16
C	A	B	A	D	D	A	B

17	18	19	20	21	22	23	24
A	C	B	D	B	A	C	D

25	26	27	28	29	30	31	32
C	C	B	D	B	A	C	D

Q1. (B) *Speed of EM waves, Lecture 1-2*

$v = 1/\sqrt{\mu_0 \varepsilon_0} \propto \varepsilon_0^{-1/2}$. Quadrupling ε_0 halves the speed, giving $c/2$.

Q2. (A) *Black body radiation, Lecture 1-2*

Every object above absolute zero contains vibrating atoms and electrons, which are accelerating charges, and accelerating charges emit electromagnetic radiation. The emission exists at every temperature above 0 K. It becomes visible as a glow only when enough of it falls in the visible range.

Q3. (D) *Black body radiation, Lecture 1-2*

The colour sequence red, orange, yellow, bluish-white means the dominant frequency rises with temperature. The observed curves also rise in height with temperature. A naive classical expectation of equal strength at all frequencies would give no peak at all.

Q4. (C) *Photons and photocurrent, Lecture 1-2*

Photon rate $= \dfrac{P}{h\nu} = \dfrac{1.0 \times 10^{-3}}{3.1 \times 1.6 \times 10^{-19}} = 2.0 \times 10^{15}$ s^{-1}. With 1% efficiency, 2.0×10^{13} electrons per second are collected, so $I = (2.0 \times 10^{13})(1.6 \times 10^{-19}) = 3.2 \times 10^{-6}$ A. Shortcut: 1.0 mW/3.1 V = 0.32 mA at 100% efficiency, and 1% of that is 3.2 μA.

Q5. (B) *Interference of single electrons, Lectures 1-2 and 4*

At a dark fringe the amplitudes from the two slits cancel, so $|\Psi|^2 = 0$. With one slit closed there is only one amplitude, which is nonzero there, so the probability becomes nonzero. Adding a second path can reduce the probability, so probabilities of separate paths do not simply add.

Q6. (B) *X-ray tube, Lecture 3*

The tube current is the number of electrons per second. Doubling it doubles the photons per second, so the intensity doubles. Each electron still gains $K = eV$ at the same voltage, so $\lambda_{\min} = hc/eV$ is unchanged.

Q7. (C) *de Broglie wavelength of electrons, Lecture 4*

$\lambda \propto 1/\sqrt{V}$. Halving λ needs four times the voltage, $4 \times 54 = 216$ V. Check: $\lambda(\text{Å}) \approx \sqrt{150/V}$ gives $\sqrt{150/54} = 1.67$ Å and $\sqrt{150/216} = 0.83$ Å.

Q8. (D) *de Broglie wavelength versus mass, Lecture 4*

At equal kinetic energy, $\lambda = h/\sqrt{2mK} \propto m^{-1/2}$. So $\lambda_p/\lambda_\alpha = \sqrt{m_\alpha/m_p} = \sqrt{4} = 2$. The charge plays no role.

Q9. (C) *Uncertainty principle, Lecture 4*

The principle is a property of waves, not of instruments. A sharply located wave packet needs a spread of wavelengths (momenta), and a sharply defined momentum needs a long wave train that is not localized. Better apparatus does not remove it.

Q10. (A) *Schrödinger equation, Lecture 5*

The time-dependent equation governs the evolution of the quantum state. Given $\psi(\vec{r}, 0)$, it fixes $\psi(\vec{r}, t)$ and hence all later probabilities $|\psi|^2$. It does not provide a Newtonian trajectory. The allowed energies come from the time-independent equation $\hat{H}\psi = E\psi$.

Q11. (B) *Energy eigenvalue equation, Lecture 5 and Chapter 5*

$\hat{H}\sin(kx) = -\dfrac{\hbar^2}{2m}\dfrac{d^2}{dx^2}\sin(kx) = \dfrac{\hbar^2 k^2}{2m}\sin(kx)$, so $E = \dfrac{(1.055\times10^{-34})^2(10^{10})^2}{2(9.11\times10^{-31})} = 6.1\times10^{-19}$ J $= 3.8$ eV. Option D comes from using h in place of $\hbar$.

Q12. (A) *Zeeman effect, Chapter 4*

Adjacent sublevels differ by $\Delta E = \mu_B B = 9.27\times10^{-24}$ J $= 5.8\times10^{-5}$ eV at 1.0 T. Option B is $2\mu_B B$, and option D is the value in joules with the wrong unit.

Q13. (D) *Orbital magnetic moment, Chapter 4*

$\mu = \dfrac{e}{2m_e}L$ with $L = \hbar\sqrt{l(l+1)} = \sqrt{2}\,\hbar$ for $l = 1$. Hence $\mu = \sqrt{2}\,\dfrac{e\hbar}{2m_e} = \sqrt{2}\,\mu_B \approx 1.41\,\mu_B$.

Q14. (D) *Selection rules, Chapter 4*

Allowed transitions need $\Delta l = \pm1$ (Δn is unrestricted). $3d \to 2p$ has $\Delta l = -1$. The transitions $2s \to 1s$ and $3p \to 2p$ have $\Delta l = 0$, and $4f \to 3s$ has $\Delta l = -3$, so all three are forbidden.

Q15. (A) *Screening and X-ray energies, Lecture 6*

Other electrons cancel part of the nuclear charge, so the effective charge is $Z_e < Z$. Level energies scale roughly as Z^2, so the $L \to K$ photon is only 10.2 eV (ultraviolet) for hydrogen but reaches the keV range for atoms with large Z. An $L \to K$ transition therefore does not always produce an X-ray.

Q16. (B) *Plum-pudding model, Chapter 4*

In this model there is no concentrated charge or mass. A heavy, fast alpha particle passing through diffuse positive charge and light electrons would be deflected only slightly. Rutherford's observation of some large-angle and backward scattering contradicted this.

Q17. (A) *Failure of the classical atom, Chapter 4*

A classical orbiting electron radiates at its orbital frequency and loses energy, so its radius and orbital frequency change continuously. The prediction is a continuous spectrum and collapse into the nucleus. Real atoms are stable and emit discrete lines.

Q18. (C) *Lyman series, Chapter 4*

$\dfrac{1}{\lambda} = R\left(1 - \dfrac{1}{n^2}\right)$. The longest wavelength is for $n = 2$: $\lambda = \dfrac{4}{3R} = \dfrac{4}{3(1.097\times10^7)} = 1.215\times10^{-7}$ m $=$ 121.5 nm. The value 91.2 nm is the series limit, 97.3 nm is the $4 \to 1$ line, and 656 nm is the Balmer H_α line.

Q19. (B) *Brackett series, Chapter 4*

$\dfrac{1}{\lambda} = R\left(\dfrac{1}{4^2} - \dfrac{1}{n^2}\right)$ with $n = 5, 6, \ldots$. The shortest wavelength is at $n \to \infty$: $\lambda = \dfrac{16}{R} = 1.459\times10^{-6}$ m $=$ 1459 nm. The value 4051 nm is the longest line ($n = 5$), 820 nm is the Paschen limit ($9/R$) and 2280 nm is the Pfund limit ($25/R$).

Q20. (D) *Standing-wave condition for Bohr orbits, Chapter 4*

$2\pi r_n = n\lambda$ gives $\lambda = \dfrac{2\pi r_n}{n} = \dfrac{2\pi n^2 a_0}{n} = 2\pi n a_0$. For $n = 2$: $\lambda = 4\pi a_0 = 4\pi(0.53) = 6.7$ Å. The orbit radius is $4a_0 = 2.1$ Å and the circumference is 13.3 Å, exactly two wavelengths.

Q21. (B) *Limitations of the Bohr model, Chapter 4*

Bohr's quantization gives $L = n\hbar$, so the ground state has $L = \hbar$. In quantum mechanics the ground state $1s$ has $l = 0$ and $L = 0$. Options A, C and D are false: the model gets the hydrogen levels and the Balmer series right. It does fail for many-electron atoms.

Q22. (A) *Bohr energies, Chapter 4*

With the Coulomb force supplying the centripetal force, $K = -E = -U/2$. For $n = 2$: $E = -13.6/4 = -3.4$ eV, so $K = +3.4$ eV and $U = -6.8$ eV.

Q23. (C) *Conditions on wavefunctions, Chapter 5*

An acceptable wavefunction is continuous, has a continuous gradient, is single-valued, and is finite everywhere, including as $|x| \to \infty$ so that it can be normalized. e^{x^2} diverges for large $|x|$. The other three are finite and normalizable.

Q24. (D) *Probabilities in a box, Chapter 5*

$P = \displaystyle\int_{L/4}^{3L/4} \frac{2}{L}\sin^2\frac{\pi x}{L}\,dx = \left[\frac{x}{L} - \frac{\sin(2\pi x/L)}{2\pi}\right]_{L/4}^{3L/4} = \frac{1}{2} + \frac{1}{\pi} \approx 0.82$. As a check, the ground state peaks at the centre, so the central half must hold well over 50% of the probability.

Q25. (C) *Harmonic oscillator wavefunctions, Chapter 5*

$\psi_n \propto H_n(\alpha x)\, e^{-\alpha x^2/2}$, and the Hermite polynomial H_n of degree n has n zeros. So $n = 3$ gives 3 nodes. (An infinite-well state with $n = 3$ has 2 nodes, which is option B.)

Q26. (C) *Particle in an infinite well, Chapter 5*

$E_1 = \dfrac{h^2}{8mL^2} = \dfrac{(6.63 \times 10^{-34})^2}{8(9.11 \times 10^{-31})(0.50 \times 10^{-9})^2} = 2.4 \times 10^{-19}$ J $= 1.5$ eV. Check: an electron in a 1.0 nm well has 0.376 eV, and halving L multiplies this by 4.

Q27. (B) *Finite well, Chapter 5*

$\delta \approx \dfrac{\hbar}{\sqrt{2m(U-E)}}$. A higher state has larger E, so $U - E$ is smaller and δ is larger. Even the ground state penetrates the walls, so option D is false.

Q28. (D) *Separation of variables, Chapter 6*

In $\psi = R(r)\Theta(\theta)\Phi(\phi)$, the potential energy depends only on r, so it appears only in the radial equation. The polar and azimuthal equations contain only angular variables and the separation constants (l and m_l).

Q29. (B) *Centrifugal term, Chapter 6*

At fixed r the term is proportional to $l(l+1)$. For $l = 2$ this is 6, for $l = 1$ it is 2, so the ratio is 3. For $l = 0$ the term vanishes.

Q30. (A) *Spectroscopic notation, Chapter 6*

The letters s, p, d, f stand for $l = 0, 1, 2, 3$, and l can only run from 0 to $n - 1$. For $n = 2$ only $l = 0, 1$ exist, so $2d$ is not allowed. States $3d$, $4f$ and $1s$ are all allowed.

Q31. (C) *Radial probability distributions, Chapter 6*

The number of radial nodes is $n - l - 1 = 3 - 0 - 1 = 2$. The radial probability $P(r) = r^2|R|^2$ has one maximum in each of the three regions separated by these two nodes, so there are $n - l = 3$ maxima.

Q32. (D) *Radiative transitions, Chapter 6*

In a single stationary state $|\Psi|^2$ does not change in time, so $\langle x \rangle$ is constant and nothing is radiated. In a superposition the cross terms vary as $e^{\pm i(E_m - E_n)t/\hbar}$, so $\langle x \rangle$ oscillates at $\nu = (E_m - E_n)/h$, the Bohr frequency of the emitted radiation.